

TOI-2046b

R-2213

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_230930 · created 2026-08-14T22:57:18+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460217.72 +/- 0.058 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.124 +/- 0.029
Transit depth (Rp/Rs)^2	1.53 +/- 0.71 [%]
Orbital Inclination (inc)	83.52 +/- 2.48
Airmass coefficient 1 (a1)	1.08 +/- 0.014
Airmass coefficient 2 (a2)	-0.051 +/- 0.038
Scatter in the residuals of the lightcurve fit is	1.33 %
Best Comparison Star	#3 - [123, 386]
Optimal Aperture	3.67
Optimal Annulus	8.0
Transit Duration (day)	0.069 +/- 0.031

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.124 ± 0.029 vs 0.1213 (deviation 0.09σ)
Duration fit vs published	0.069 d vs 0.1033 d (deviation 1.11σ)
Mid-time O–C	3.2 min
Depth SNR	4.3
Residual scatter	1.33 %

Light curve

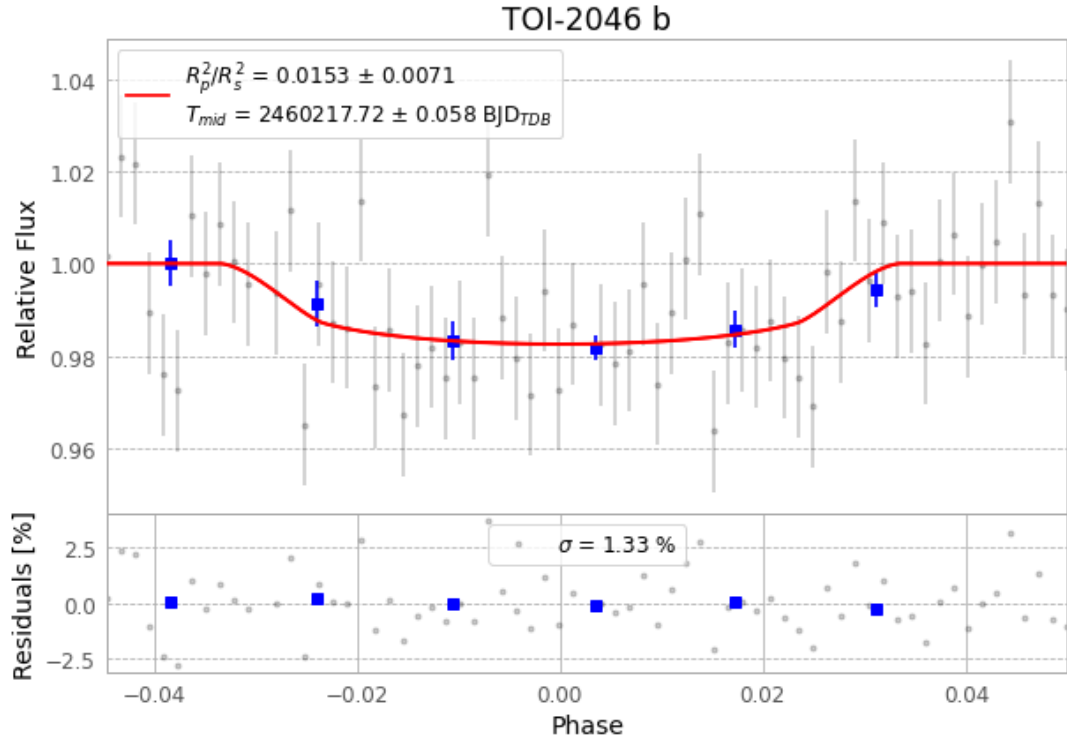


Figure 1 — FinalLightCurve_TOI-2046 b_2023-09-29.png (SHA-256 150119fc90bc5393...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 241], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 07e9ae0ab655a593...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), TOI2046230930033615.FITS → TOI2046230930070015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-230930.log (d0399ee7a9c8...), FinalLightCurve_TOI-2046 b_2023-09-29.png (150119fc90bc...), FinalLightCurve_TOI-2046 b_2023-09-29.csv (c56598b9672a...)

Compute resources

Wall time 82.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 22:57Z.